Duc-Cuong Vu

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Education

Master of Sci. in Automation and Control, *Hanoi University of Science and Technology* (*HUST*), *Jul 2024 - present*
Funded by: *Master, PhD Scholarship Programme of Vingroup Innovation Foundation* (*VinIF*)

Bachelor of Sci. in Automation and Control, *Hanoi University of Science and Technology* (*HUST*), *Oct 2020 - Mar 2024*
Excellent degree, GPA: 3.71/4 (Rank: 22/499, top 5%). Finished the 4-year BSc program in just 3.5 years.

Selected publications

- ⋆ Glocal trajectory generation and tracking control for AUVs with optimal coverage sensor networks
  Ocean Engineering (*SCIE Q1*) (*2025*) [pdf]
  D.C. Vu†, *S. Tran*†, *T.L. Nguyen, and D.C. Hoang* †: *equally contribution*
- ⋆ Lagrangian-based modeling and safety-critical controls for Stewart platforms under marine operations
  Ocean Engineering (*SCIE Q1*) (*2025*) [pdf]
  D.C. Vu, *D.H. Nguyen, M.N. Vu, and T.L. Nguyen*
- ⋆ CBFs-based Model Predictive Control for Obstacle Avoidance with Tilt Angle Limitation for Ball-Balancing Robots
  IEEE Acess (*SCIE Q2*) (*2025*) [pdf]
  M.D. Pham, ***D.C. Vu****, T.T.H. Nguyen, T.V.A. Nguyen, M.N. Vu, and T.L. Nguyen*
- ⋆ A novel approach of Consensus-based Finite-time DSMC for Stewart platform manipulators motion tracking
  Results in Engineering (*ESCI Q1*) (*2025*) [pdf]
  D.C. Vu*, D.H. Nguyen, and T.L. Nguyen*
- ⋆ Unifying Hierarchical Sliding Mode Control and Control Barrier Function for Tilt Angle Constraint of a Ball-Balancing Robot
  IET Cyber-Systems and Robotics (*SCIE Q3*) (*2025*) [pdf]
  T.T.H. Nguyen, ***D.C. Vu****, M.D. Pham, T.V.A. Nguyen, and T.L. Nguyen*
- ⋆ Time-optimal trajectory generation and observer-based hierarchical sliding mode control for ballbots with system constraints
  International Journal of Robust and Nonlinear Control - IJRNC (*SCIE Q1*) (*2024*) [pdf]
  D.C. Vu*, M.D. Pham, T.T.H. Nguyen, T.V.A. Nguyen, and T.L. Nguyen*

Other manuscripts

- ⋆ Decentralized Control Strategy with Consistent Joint-to-Actuator Control Law for Parallel Mechanisms on a Humanoid Platform
  Submitted manuscript [pdf]
  T.V. Trung, V.T. Nguyen, ***D.C. Vu****, D.H. Nguyen, M.C. Nguyen, and M.N. Vu*
- ⋆ Adaptive Whole-Body MPC with Optimal Posture Planning for Humanoid Robots under Unknown Payloads
  Submitted manuscript [pdf]
  M.C. Nguyen†, ***D.C. Vu***†, *V.A. Vu, T. Nguyen, D.T. Pham, M.H. Phung, A.V. Ngo, and M.N. Vu* †: *equally contribution*
- ⋆ Conformal-Seeded Hybrid Inverse Kinematics for Offset Redundant Manipulators
  Submitted manuscript [pdf]
  D.C. Vu*, V.T. Nguyen, D.H. Nguyen, M.C. Nguyen, T.V. Trung, A. Kugi, and M.N. Vu*

Experience

Robotics Engineer at VinRobotics *Sep 2025 - present*
Full-time, supervised by PhD. Minh Nhat Vu *Hanoi, Vietnam*

- ⋆ Conduct pipeline for robot model standardization, kinematics calibration, system identification, and advanced control algorithm for industrial robots.
- ⋆ Design motion planning and motion control for VinRobotics humanoids robot, conduct scientific papers which are submitted to IROS2026 and IEEE Transaction on Mechatronics.

Research project assistant at HUST *Jan 2025 - present*

Contract (Hybrid) supervised by PhD. Chinh Hoang Duc (PI) and Assoc.Prof.PhD. Tung Lam Nguyen, *Hanoi, Vietnam*

- ⋆ Work under project: Robot navigation system integrating sensor network and wireless communication.
- ⋆ Designed and developed a comprehensive MuJoCo-based simulation environment for AUVs, incorporating underwater dynamics, sensor feedback, environmental disturbances, and communication constraints to evaluate system performance.
- ⋆ Implemented and validated advanced control algorithms for navigation, obstacle avoidance, and trajectory tracking, while collaborating on integration, troubleshooting, and authoring a peer-reviewed scientific paper.

Research project assistant at HUST *Jan 2025 - Oct 2025 (10 mos)*

Contract (Hybrid) supervised by PhD. Minh Nhat Vu (PI) and Assoc.Prof.PhD. Tung Lam Nguyen, *Hanoi, Vietnam*

- ⋆ Work under project: Advanced Control of a Ship-Mounted Stewart Platform for Marine Applications
- ⋆ Funded by KIST Korea Institute of Science and Technology
- ⋆ Designing and implementing advanced control algorithms for the Stewart platform, including safety-critical and robust control strategies tailored for marine environments.
- ⋆ Implemented and validated advanced control algorithms for navigation, obstacle avoidance, and trajectory tracking, while collaborating on integration, troubleshooting, and authoring a peer-reviewed scientific paper.

Graduate Student Research at HUST *May 2024 - present*

Part-time (Hybrid) supervised by Assoc.Prof.PhD. Tung Lam Nguyen, *Hanoi, Vietnam*

- ⋆ Work under master project: Design control structures for Parallel Platforms in Maritime applications.
- ⋆ Funded by VinIF.
- ⋆ Designed and implemented advanced control algorithms for a Stewart platform in marine environments, supported by high-fidelity simulations (Simscape, MuJoCo) and validated through a full experimental setup (mechanical assembly, hardware integration, Linux real-time kernel, EtherCAT communication).
- ⋆ Collaborated with cross-institutional teams on system integration, troubleshooting, and documentation, while authoring peer-reviewed publications and presenting outcomes to academic and industrial partners.

Student Intern at HUST *Oct 2021 - Apr 2024 (2 yrs 7 mos)*

Internship (Hybrid) supervised by Assoc.Prof.PhD. Tung Lam Nguyen, *Hanoi, Vietnam*

- ⋆ Work under bachelor project: Balancing, motion planning, and tracking control for ballbot systems.
- ⋆ Developed mathematical models and simulation environments for 3D ballbot systems, focusing on nonlinear dynamics, trajectory generation, and safety constraints.
- ⋆ Conducted research on modeling and simulation, advanced control strategies, and practical implementation for the Ball-Balancing Robot.
- ⋆ Authored and co-authored peer-reviewed journal papers based on the project outcomes, including publications in the International Journal of Robust and Nonlinear Control (RNC) and IEEE Access.

Academic activities

Invited review for

- ⋆ *Nonlinear Dynamics*
- ⋆ *International Conference on Intelligent Robots and Systems (IROS, 2026)*
- ⋆ *IEEE Robotics and Automation Letters (RA-L)*

Teaching, Seminars and Talks

2025: Talk "MuJoCo for Advanced Physics Simulation: From manipulators to autonomous vehicles" for "Motion Control" master course at HUST and MoCAR seminar [pdf]

2025: Seminar "Underwater Vehicles" for modeling training of Autonomous Underwarter Vehicle at MEG-MoCAR [pdf]

Honours & awards

Master, PhD Scholarship Programme *Vingroup Innovation Foundation (VINIF)*

Best Thesis Defense Award *Hanoi University of Science and Technology*

Skills

Programming	Python, C/C++, MATLAB.
Engineering Tools	Git, ROS2, IgH EtherCAT, real-time Linux, Acados, CasADI, Pinocchio, MuJoCo, Simulink, Simscape, SolidWorks, Motion Capture system, PCB design, etc.
Research	Scientific writing, publications, presentations, literature analysis, experimental validation.

Languages

Vietnamese	Mother tongue.
English	IELTS Academic: Overall 6.5 (L:6.0, R:8.0, W:6.5, S:5.5)